ANTIBIOTICS

Complete Pharmacology Exam Summary

1 β-Lactams, Vancomycin & Metronidazole

2 Aminoglycosides

3 Sulfonamides, Trimethoprim & Co-trimoxazole

4 Broad-spectrum antibiotics
(Macrolides, Lincosamides, Tetracyclines, Chloramphenicol …)

5 Antimycobacterial drugs (TB & Leprosy)

6 Exam cheat-sheet & self-test

ملخص شامل ودقيق للمحاضرات الخمس — مصطلحات إنجليزية كما في الامتحان + شرح بالعربية

Contents

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1	β-Lactams, Vancomycin & Metronidazole — penicillins, cephalosporins, carbapenems, monobactam
2	Aminoglycosides — MOA, toxicity, each drug, renal dose adjustment
3	Sulfonamides, Trimethoprim, Co-trimoxazole — folate pathway
4	Broad-spectrum antibiotics — macrolides, lincosamides, fusidic acid, polymyxins, mupirocin, bacitracin, tetracyclines, chloramphenicol
5	Antimycobacterial drugs — TB regimens & leprosy regimens
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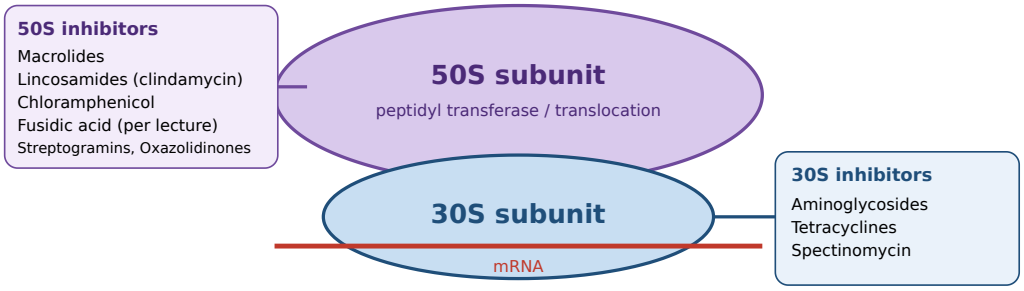
كيف تذاكر من هذا الملف؟

- اقرأ كل جزء ثم اختبر نفسك بالأسئلة السريعة في الجزء **6**.
- الصناديق **الحمراء** = نقاط يتكرر السؤال عنها. الصناديق **الخضراء** = طرق حفظ (Mnemonics). الصناديق **الزرقاء** = تنبيه على أخطاء/غموض في شرائح المحاضرات.
- كل المعلومات مأخوذة من المحاضرات الخمس المرفوعة؛ أي إضافة من خارجها موسومة بعلامة **(+)**.

0 · Big Picture

Where each antibiotic group acts | مكان عمل كل مجموعة

Ribosome map (70S = 30S + 50S)



(+) Mnemonic: "AT 30, CELL at 50"
Aminoglycosides & Tetracyclines at 30S · Chloramphenicol, Erythromycin, Lincosamides, Linezolid at 50S

Figure 0.1 — drawn from the lecture figure "70S ribosome"

شرح

الريبوسوم البكتيري **70S** يتكون من وحدتين: **30S** (تستقبل mRNA وتبدأ التخليق) و**50S** (تجري ربط الأحماض الأمينية peptidyl transferase والانتقال translocation). **Aminoglycosides** و **Macrolides, Tetracyclines** → 30S | **Lincosamides, Chloramphenicol, Fusidic acid** → 50S.

Mechanism map

Target	Drugs
Cell-wall synthesis	β-lactams (penicillins, cephalosporins, carbapenems, monobactam) · Vancomycin (binds D-Ala-D-Ala) · Bacitracin · (TB: isoniazid, pyrazinamide, ethionamide; cycloserine blocks alanine racemase)
30S ribosome	Aminoglycosides · Tetracyclines
50S ribosome	Macrolides · Telithromycin · Lincosamides · Chloramphenicol · Fusidic acid
Folate pathway	Sulfonamides, Dapsone, PAS (block DHPS) · Trimethoprim, Pyrimethamine (block DHFR)
RNA synthesis	Rifampin (DNA-dependent RNA polymerase)
DNA damage / replication	Metronidazole (free radicals → strand breaks) · Clofazimine (DNA replication)
Cell membrane	Polymyxin B & Colistin (cationic detergents)
tRNA synthetase	Mupirocin (isoleucyl-tRNA synthetase)

Bactericidal vs bacteriostatic

CIDAL Bactericidal	STATIC Bacteriostatic
β-lactams · Aminoglycosides · Vancomycin · Polymyxins · Bacitracin · Rifampin · Pyrazinamide · Capreomycin · Clofazimine · Metronidazole · **Co-trimoxazole** (the combination)	Sulfonamides · Dapsone · Tetracyclines · Lincosamides · Ethambutol · PAS · Cycloserine · **Chloramphenicol** (cidal at high conc.) · **Macrolides** (cidal or static depending on conc./organism/pH) · **Isoniazid** (static on resting, cidal on dividing bacilli)

Exam point: Penicillins should not be combined with a bacteriostatic antibiotic — growth inhibition reduces penicillin effectiveness (penicillins kill only *actively growing* cells).

1 · β-Lactams, Vancomycin & Metronidazole

Source: "Inhibitors of Cell Wall Synthesis (β-Lactam antibiotics and Vancomycin)" — Dr Eman Maddy

1.1 Classification

β-lactams = antibiotics containing a **β-lactam ring** (crucial for the mechanism: disrupting bacterial cell-wall synthesis).

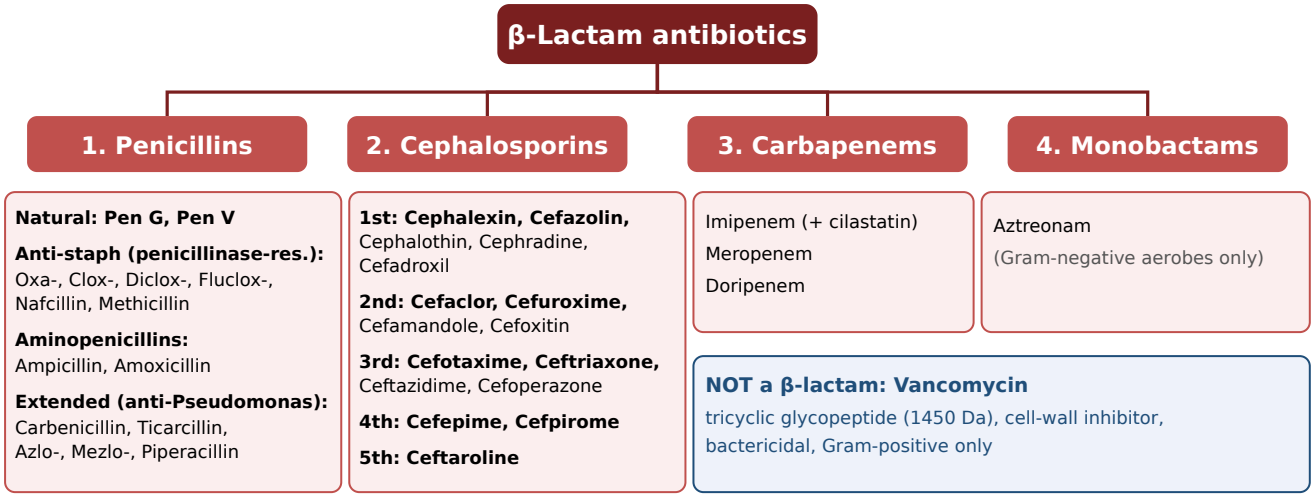


Figure 1.1 — β-lactam family tree

1.2 Mechanism of action

- All β-lactams interfere with **bacterial cell-wall synthesis**. The wall = peptidoglycan strands of alternating **NAG** (N-acetyl-glucosamine) and **NAM** (N-acetyl-muramic acid); NAM carries short peptide chains that are **cross-linked** — the **last step**, done by the enzyme **transpeptidase**.
- Penicillins (and cephalosporins) bind to and inhibit **transpeptidase** → no cross-linking → no wall formation.
- **PBPs (Penicillin-Binding Proteins)** = bacterial membrane enzymes involved in wall synthesis; penicillins inactivate them → morphological changes or **lysis**.
- Also: removal/inactivation of an inhibitor of **autolytic enzymes** → lysis.
- Bactericidal **only if active wall synthesis occurs** — active against growing, not resting cells.

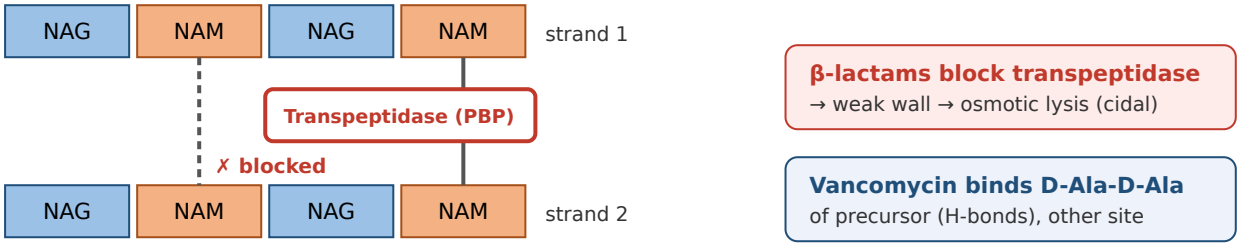


Figure 1.2 — last step of cell-wall synthesis and where drugs act

شرح

جدار البكتيريا مبني من سلاسل peptidoglycan متشابكة بإنزيم **transpeptidase** (هو نفسه الـ PBP). البنسلين يعطّل هذا الإنزيم فيضعف الجدار وتنفجر البكتيريا. لذلك يقتل فقط البكتيريا **النشطة النمو**، ولا يُعطى مع مضاد **bacteriostatic** (لأنه يوقف النمو فيبطل مفعول البنسلين).

1.3 Resistance & β-lactamase inhibitors

Mechanism	Detail
1. Enzymatic cleavage	β-lactamases (penicillinases, cephalosporinases, carbapenemases) hydrolyze the amide bond of the β-lactam ring
2. Target (PBP) alteration	Low-affinity PBPs: **mecA** gene → **PBP2a in MRSA**; mutated PBP in *S. pneumoniae*
3. Porin-channel loss	Down-regulation/mutation of outer-membrane porins in Gram-negatives (e.g. **OprD loss in Pseudomonas**)
4. Active efflux pumps	Over-expressed multidrug efflux pumps pump drug out of the periplasmic space

β-lactamase inhibitor	Combined with	Brand
Clavulanic acid	Amoxicillin (**co-amoxiclav**) or Ticarcillin	Augmentin®
Sulbactam	Ampicillin	Unasyn®
Tazobactam	Piperacillin	Zosyn®

1.4 Penicillins — Pharmacokinetics

- **Absorption (oral):** varies with acid stability and food; some poorly absorbed because of acid lability — **Pen G**. Food delays/decreases absorption of some (e.g. **ampicillin**) → give **≥1 h before or 2 h after meals**. Parenteral: IM, IV, SC.
- **Intrathecal injection is NOT recommended, especially Pen G — may cause convulsions.**
- **Distribution:** widely distributed but do **not enter mammalian cells**; do **not cross BBB** unless meninges are **inflamed** (then therapeutic levels reach CSF).
- **Excretion:** mostly rapid renal excretion, mainly by **tubular secretion**; partly in bile.

1.5 Natural penicillins

Penicillin G (Benzylpenicillin)

- First penicillin isolated; used as stable Na or K salts (crystalline penicillin); expressed in **IU**.
- **60–70 % destroyed by gastric acid** → given IV or IM.
- **t½ ≈ 30 min** after IM (main clinical problem) → prolonged by slow-release forms: **Procaine penicillin G** and **Benzathine penicillin**.

Clinical uses of Pen G

Infection	Notes
Streptococcal	Pharyngitis, tonsillitis, otitis media, scarlet fever, rheumatic fever, endocarditis
Pneumococcal	Pneumonia, meningitis, arthritis, osteomyelitis, endocarditis, pericarditis, peritonitis
Meningococcal	Meningitis
Gonococcal	Resistance ↑ → **Ceftriaxone 250 mg IM single dose** (DOC), or Procaine Pen G 4.8 million units + **Probenecid 1 g** orally
Syphilis	Pen G or Procaine Pen G
Diphtheria	Specific **antitoxin** is the only effective treatment; Pen G eliminates the **carrier state**
Tetanus & gas gangrene	Pen G kills the causative organism
Anthrax	**Pen G = drug of choice**
Actinomycosis	Pen G usually effective
Gingivostomatitis	Low-dose Pen V (500 mg q6h × 7 days)
Rat-bite fever	**Pen G = DOC**
Listeria	Meningitis — large doses of Pen G
Staphylococcal	Most staph make penicillinase → use penicillinase-resistant penicillin
Lyme disease	e.g. Lyme arthritis

Penicillin V (Phenoxymethylpenicillin)

- **Acid-stable → oral**. Spectrum ≈ Pen G, but **Pen G is more active against Gram-negatives**.
- Oral Pen V only for **mild infections**: respiratory infections in children, pharyngitis, otitis media, sinusitis.

Prophylactic uses of penicillins

- Streptococcal disease in closed populations (schools, military); deep burns (Pen G/V).
- Recurrences of **rheumatic fever** (Pen V or G).
- Sexual contacts of gonorrhea/syphilis patients.
- Valvular heart disease patients undergoing dental extraction, tonsillectomy, other bacteremia-causing procedures (prevent endocarditis).

1.6 Semisynthetic penicillins

Made by adding different side chains (at R) to the penicillin nucleus. Three groups:

Group	Drugs	Key points
A. Penicillinase-resistant (anti-staph)	Oxacillin (IV), Cloxacillin (PO), Dicloxacillin (PO), Flucloxacillin, Nafcillin (IV), Methicillin	Resist staphylococcal penicillinase. **Only indication: penicillinase-producing staph** (RTI, enterocolitis, osteomyelitis). Less active than Pen G on other organisms. Well absorbed orally (**except Methicillin**); food interferes → give 1–2 h from meals. **Methicillin**: acid-labile → injection only; **nephrotoxic** → rarely used.
B. Broad-spectrum (aminopenicillins)	Ampicillin (PO), Amoxicillin; prodrugs: Pivampicillin, Talampicillin, Bacampicillin	↑ activity vs **Gram-negatives** than Pen G. **All destroyed by β-lactamase** → ineffective for most staph. Prodrugs release ampicillin in liver: better absorbed, **less diarrhea**. Ampicillin: acid stable, moderately absorbed, food delays absorption.
C. Extended-spectrum (anti-Pseudomonas)	Carbenicillin (IV), Ticarcillin (IV), Azlocillin, Mezlocillin, Piperacillin	**Not absorbed orally** (IM/IV). Broad Gram+, Gram–, anaerobic cover + **Pseudomonas**. **Inactivated by β-lactamases.**

Ampicillin — indications

- URTI (sinusitis, otitis media, acute/chronic bronchitis) — ampicillin & amoxicillin.
- Uncomplicated UTI.
- Meningitis in **children**: ampicillin + 3rd-generation cephalosporin.
- **Salmonella** (typhoid fever): high doses.

Amoxicillin

- Similar to ampicillin; **food does NOT interfere** with absorption; **less diarrhea**.
- With clavulanic acid = **co-amoxiclav (Augmentin)**.
- Uses: URTI (sinusitis, tonsillitis, otitis media); LRTI (bronchitis, bronchopneumonia, lung abscess); GU (cystitis, urethritis, pyelonephritis).

Extended-spectrum — clinical indications

- *P. aeruginosa* infections (nosocomial pneumonia, febrile neutropenia, complicated UTI) · intra-abdominal sepsis (polymicrobial peritonitis) · severe diabetic foot infections.

1.7 Penicillin side-effects

- Least direct toxicity of all antibiotics. Main problem = **hypersensitivity**: degradation products (esp. **penicilloic acid**) combine with host proteins → antigenic. **Cross-allergy between all penicillins.**
- Common: skin rash, urticaria, fever. Rare but serious: **anaphylactic shock** (circulatory collapse, bronchospasm/edema) — **Adrenaline is the therapy of choice**. Delayed reaction after 1–2 weeks (mild fever, rash, urticaria).
- Always take allergy history; intradermal test may help but can itself cause fatal anaphylaxis.
- Others: GI upset if oral; pain at injection site; **neurotoxicity** (confusion, convulsions, coma) with large IV doses, especially in **renal insufficiency**; **ampicillin-induced pseudomembranous colitis**; **carbenicillin → hypokalemia**.

1.8 Cephalosporins

Semisynthetic, derived from **Cephalosporin C** (fungus *Cephalosporium*); structure like penicillins; side chains added at R1 and R2. **Bactericidal**, same MOA as penicillins (cell-wall synthesis inhibition).

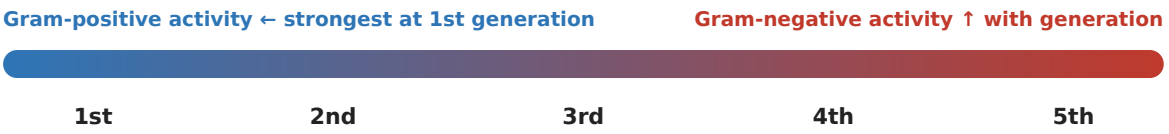


Figure 1.3 — cephalosporin generations trend

Gen.	Drugs	Spectrum & key facts
1st	Cephalothin, Cephalexin, Cefazolin, Cephradine, Cefadroxil	Good vs **Gram +ve**, modest vs Gram –ve. **Cephalothin**: good vs *E. coli, Klebsiella, pneumoniae*; **resistant to staph β-lactamase**; very effective in severe staph infection (endocarditis); not absorbed orally → IV (IM very painful). **Cephalexin**: oral; less active vs β-lactamase-producing staph; used for **UTI** and minor respiratory infections (bronchitis). Also used for **surgical prophylaxis**.
2nd	Cefaclor, Cefuroxime, Cefamandole, Cefoxitin (cephamycin)	↓ Gram+ and ↑ Gram– activity vs 1st gen. **Cefaclor**: oral; sinusitis and otitis media in patients **allergic to ampicillin**. **Cefuroxime**: IM and oral; β-lactamase resistant; active vs **H. influenzae, N. gonorrhoeae** (not Pseudomonas); uses: CAP, otitis media, sinusitis, **meningitis (crosses BBB)**, penicillin-resistant gonorrhoea.
3rd	Cefotaxime, Ceftriaxone, Ceftazidime, Cefoperazone	Extended Gram– spectrum (some inhibit **Pseudomonas**); less active vs Gram+ than 1st; β-lactamase resistant; **none effective orally**. **Cefotaxime**: Gram– meningitis, septicemia. **Ceftriaxone**: bacterial meningitis, severe CAP, **gonorrhea (DOC)**, Lyme disease. **Ceftazidime**: good anti-**Pseudomonas**.
4th	Cefepime, Cefpirome	Broader than 3rd; more β-lactamase-resistant; for aerobic Gram– resistant to 3rd gen. **Dual spectrum**: Gram+ activity (MSSA, streptococci) + anti-Pseudomonal activity like ceftazidime. Empiric: febrile neutropenia, hospital-acquired pneumonia, healthcare-associated intra-abdominal infections.
5th	Ceftaroline (prodrug **ceftaroline fosamil**, IV)	Broad Gram+ activity + Gram– activity like ceftriaxone. Indicated for **complicated skin infections** and **CAP**.

Clinical uses of cephalosporins (lecture list)

- Alternative to penicillins in patients who cannot tolerate them (strep & staph infections — cephalothin).
- Prophylaxis before/after surgery (1st generation).
- Gram-negative **meningitis** (cefuroxime, cefotaxime).
- Gonorrhoea by β-lactamase producers — **ceftriaxone = DOC** (alternative to Pen G or cefuroxime).
- Mixed aerobic/anaerobic infections of chest, abdomen, pelvis (3rd gen) and septicemia.

Side-effects of cephalosporins

- Hypersensitivity like penicillins; **cross-reaction ≈ 10 %** between cephalosporin and penicillin allergy.
- **Nephrotoxicity**.
- **Disulfiram-like reaction** with alcohol.
- Diarrhea (oral cephalosporins; cefoperazone — biliary excretion).
- **Hypoprothrombinemia & bleeding** → prevented by **vitamin K**.

1.9 Carbapenems & Monobactams

Class	Details
Carbapenems	Imipenem (IV), Meropenem (IV), Doripenem. **Broadest spectrum of all β-lactams** (aerobic Gram+ and Gram–, anaerobes). Resistant to all β-lactamases (some organisms now resistant). **Imipenem is inactivated by renal dehydropeptidase → always given with cilastatin** (inhibitor). **Uses:** infections by organisms resistant to other drugs (e.g. *P. aeruginosa*); mixed aerobic/anaerobic infections; **Enterobacter** (resist its β-lactamase); **DOC for ESBL-producing Gram– bacteria**; polymicrobial intra-abdominal sepsis, necrotizing soft-tissue infection; empiric treatment of MDR nosocomial pneumonia, febrile neutropenia, complicated pyelonephritis. **Side-effects:** as other β-lactams + **nausea and vomiting** (most frequent); **neurotoxicity/seizures** with high plasma levels (imipenem in renal failure); meropenem & doripenem much less likely. Penicillin-allergic patients may also be allergic to carbapenems.
Monobactam Aztreonam	Active **only vs aerobic Gram-negative** bacteria. Routes: IV, IM, **inhaled aerosol** (Cayston: chronic *Pseudomonas* colonization in **cystic fibrosis**). Resistant to most β-lactamases. **Little cross-reactivity** with other β-lactams → **safe alternative in penicillin/cephalosporin/carbapenem-allergic** patients.

1.10 Vancomycin (NOT a β-lactam)

- Tricyclic **glycopeptide** (MW ≈ 1450 Da) from *Amycolatopsis orientalis*; **bactericidal**; inhibits cell-wall synthesis by binding via H-bonds to the **D-alanyl-D-alanine** terminus of peptidoglycan precursors.
- **Spectrum — Gram-positive only: MRSA & MRSE**, *S. pneumoniae* (incl. penicillin-resistant), *Enterococcus faecalis/faecium* (sensitive strains), *C. difficile* (oral).
- **Not absorbed orally.** IV for systemic infection: severe MRSA (bacteremia, endocarditis, osteomyelitis, pneumonia), prosthetic joint/vascular graft infections, empirical bacterial meningitis (with ceftriaxone + ampicillin).
- **Oral** vancomycin: **only** for *C. difficile* colitis (pseudomembranous). **IV vancomycin is ineffective for C. difficile** (doesn't reach lumen).

Vancomycin side-effects

- Thrombophlebitis.
- **Ototoxicity & nephrotoxicity** (dose-dependent AKI/tubular necrosis); ↑ risk with aminoglycosides, piperacillin-tazobactam, loop diuretics.
- **Red man syndrome** — non-immunologic direct mast-cell degranulation from rapid IV infusion → histamine → flushing, erythema of face/neck, hypotension. **Prevention: slow IV infusion (~60 min) + antihistamines.**

1.11 Metronidazole

- Antibiotic + antiprotozoal for **anaerobic bacterial** and parasitic infections.
- **MOA:** diffuses into organism → **cytotoxic free-radical generation** → interacts with DNA → loss of helical structure and strand breakage → cell death.
- **Indications:** anaerobes (*Bacteroides fragilis, Clostridium* below the diaphragm) · *H. pylori* (triple/quadruple therapy) · *Gardnerella vaginalis* (bacterial vaginosis) · *Giardia lamblia* (watery diarrhea) · *Trichomonas vaginalis* (vaginitis) · intra-abdominal peritonitis/abscess (+ ceftriaxone for aerobes) · mild–moderate *C. difficile* colitis (second-line to oral vancomycin) · aspiration pneumonia/lung abscess (oral anaerobes: *Prevotella, Fusobacterium*).
- **Adverse effects:** nausea, vomiting, diarrhea, bitter **metallic taste**; **disulfiram-like reaction** with alcohol (inhibits aldehyde dehydrogenase: cramps, vomiting, flushing, tachycardia, headache); **neurotoxicity** (peripheral neuropathy, cerebellar ataxia, encephalopathy, seizures with prolonged high doses).
- Counsel: **complete the entire course** even if symptoms improve.

أهم ما يُسأل عنه في هذا الجزء

- **Imipenem + cilastatin** (لأن dehydropeptidase في الكلية يدمّره) — والـ**seizures** مع imipenem خصوصًا في الفشل الكلوي.
- **Aztreonam**: Gram-negative aerobes فقط، وآمن في حساسية البنسلين.
- **Vancomycin**: يُعطى IV لـ MRSA، وفمويًا فقط لـ C. difficile؛ و**Red man syndrome** سببه سرعة التسريب (ليست حساسية حقيقية).
- **Ceftriaxone** = DOC للسيلان (gonorrhea)؛ **cefuroxime** يعبر BBB؛ **ceftazidime** ضد Pseudomonas.
- الحساسية المتصالبة بين البنسلين والسيفالوسبورين ≈ **%10**؛ ومضاد الحساسية الفوري: **Adrenaline**.
- Cephalosporins: **hypoprothrombinemia** (vitamin K) و**disulfiram-like reaction**.

(+) Mnemonic — Anti-staph penicillins: "Nafcillin, Oxacillin, Dicloxacillin, Cloxacillin, Flucloxacillin, Methicillin" = all end in **-cillin** and start with N/O/D/C/F/M → "*NO DC FM*".

Lecture notes: The slides list Cefoperazone under 3rd generation and Cefoxitin as a cephamycin within 2nd generation (as written). The slide states Methicillin is "nephrotoxic, rarely used". The line "Pen G more active against Gram-negative than Pen V" is as in the slides.

2 · Aminoglycoside Antibiotics

Source: "Aminoglycoside antibiotics" lecture

2.1 Introduction

- **Streptomycin — 1944**, from the actinomycete *Streptomyces griseus* (first aminoglycoside).
- **Bactericidal**; interfere with **protein synthesis**; used for **aerobic Gram –ve** bacteria.
- All resemble each other in MOA, pharmacokinetic, therapeutic and toxic properties.
- **Relatively low margin of safety** → **ototoxicity** and **nephrotoxicity**.

Chemistry

General aminoglycoside structure

Amino sugar **–O–** 2-deoxystreptamine **–O–** Amino sugar

Streptomycin

Streptidine **–O–** Streptose (amino sugar) **–O–** N-methyl-L-glucosamine

Streptobiosamine = streptose + N-methyl-L-glucosamine

2.2 Classification

Systemic	Topical
Streptomycin · Gentamicin · **Kanamycin** · **Amikacin** · Sisomicin · **Tobramycin** · Netilmicin	**Neomycin · Framycetin**

(Red = highlighted in the lecture slide.)

2.3 Mechanism of action

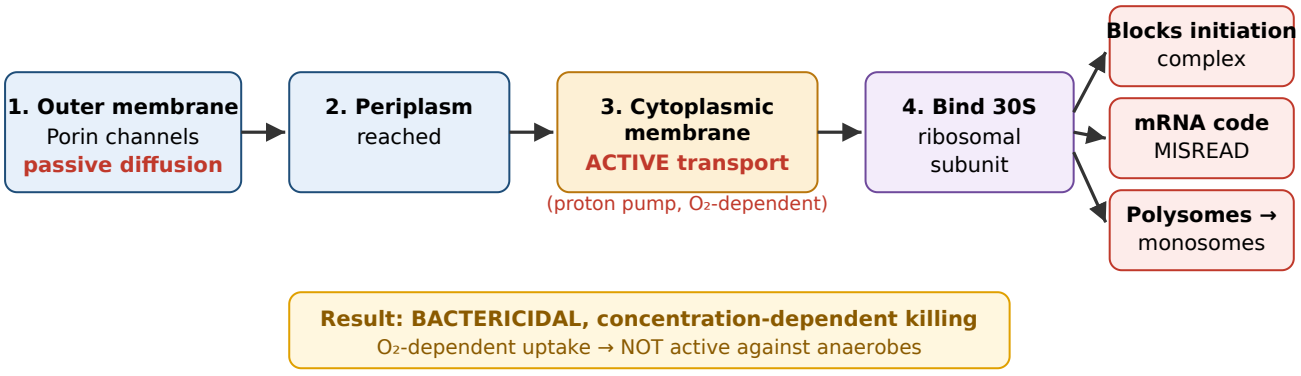


Figure 2.1 — uptake and action of aminoglycosides

شرح

الدواء يدخل أولًا عبر قنوات الـ **porin** بالانتشار السلبي، ثم يُنقل عبر الغشاء السيتوبلازمي **بنقل نشط يعتمد على الأكسجين** (لذلك لا يؤثر على اللاهوائيات). بعدها يرتبط بـ **30S** فيمنع تكوين initiation complex ويسبب **قراءة خاطئة** للشيفرة على mRNA وتفكك polysomes إلى monosomes ← موت البكتيريا.

2.4 Post-antibiotic effect & dosing

- Aminoglycosides show **concentration-dependent killing** and a significant **post-antibiotic effect (PAE)**.
- **Single daily dosing** is at least as effective as, and no more toxic than, multiple dosing.

2.5 Mechanisms of resistance

- Plasmid-mediated bacterial **transferase enzymes** inactivate aminoglycosides.
- ↓ transport into the bacterial cytosol.
- Deletion/alteration of the **receptor protein on the 30S** subunit (mutation) → prevents attachment. (Streptomycin: ribosomal resistance develops fast.)

2.6 Antibacterial spectrum

Susceptible	Examples
Primarily Gram –ve aerobic bacilli	**P**roteus, **P**seudomonas · **E**. coli, Enterobacter · **K**lebsiella · **S**higella (slide highlights the letters P-C-K-S)
Only a few Gram +ve cocci	*Staph. aureus*, *Strep. viridans*
NOT effective against	Gram +ve bacilli · Gram –ve cocci · **anaerobes**

(+) Mnemonic: "**PECKS**" — **P**roteus/**P**seudomonas, **E**.coli/Enterobacter, **K**lebsiella, **S**higella.

2.7 Pharmacokinetics

- Highly polar, basic drugs → **poor oral bioavailability**; given **parenterally (IV/IM)** or applied locally.
- Poorly distributed, poorly protein bound; **do not penetrate the CNS**.
- No significant metabolism; nearly all IV dose **excreted unchanged in urine**.
- **Dose adjustment is needed in renal insufficiency.**

Dose adjustment in renal insufficiency

Formulas (as in slides)
Dose in renal insufficiency = **Normal therapeutic dose ÷ serum creatinine (mg/dl)**

Cockcroft–Gault:
CrCl = [(140 − age) × weight(kg)] ÷ (sCr × 72)
Females × 0.85

Corrected dose = **Normal dose × patient CrCl ÷ Normal CrCl**

CrCl (ml/min)	% of max daily dose	Frequency
100	100	Every 24 h
75	75	
50	50	
25	25	
20	80	Every 48 h
10	60	
< 10	40	

2.8 Clinical uses

- **Gram –ve bacillary infection:** septicaemia, pelvic & abdominal sepsis.
- **Bacterial endocarditis** — enterococcal, streptococcal or staphylococcal infection of heart valves.
- Pneumonias, **tuberculosis**; **tularemia**; **plague**; **brucellosis**.
- **Topical** (neomycin, framycetin): infections of conjunctiva or external ear.
- To **sterilize the bowel** in patients on immunosuppressive therapy, before surgery, and in **hepatic coma**.

2.9 Shared toxicities

1. OTOTOXICITY
VIII cranial nerve
cochlear + vestibular

2. NEPHROTOXICITY
↓ GFR, ↑ serum creatinine
reversible if stopped early

3. NEUROMUSCULAR BLOCKADE
↑ myasthenic weakness

Toxicity	Details
Ototoxicity	Impairment of **VIII cranial nerve**; **may be irreversible**. **Cochlear damage** (hearing loss, tinnitus): more with **neomycin, amikacin, kanamycin**. **Vestibular damage** (vertigo, ataxia, loss of balance): more with **streptomycin, gentamicin**. **Tobramycin**: both types. **Netilmicin**: claimed low ototoxicity.
Nephrotoxicity	**Gentamicin, amikacin, tobramycin** more toxic than streptomycin. Responsible for **10–15 %** of all renal-failure cases. **Reversible** if drug promptly discontinued. ↓GFR, ↑ sr creatinine. ↓ clearance of antibiotic → ↑ ototoxicity.
Neuromuscular blockade	By (1) **displacing Ca^{2+}** from the NM junction, (2) blocking post-synaptic **N_M receptors**, (3) **inhibiting ACh release** from motor nerve. **Neomycin & streptomycin**: more propensity; **tobramycin least likely**. Myasthenic weakness ↑ by these drugs.

Precautions / contraindications

- **Pregnancy** (foetal ototoxicity) · **elderly** · kidney disease.
- With other **ototoxic** drugs (**furosemide, minocycline**) or **nephrotoxic** drugs (**vancomycin, cisplatin**).
- Cautious use of **muscle relaxants**.
- **Do not mix with any other drug in the same syringe.**

2.10 Individual drugs

Drug	Key facts
Streptomycin	Ribosomal resistance develops fast → limited as single agent. **Plague, tularemia, brucellosis** (in combination with **tetracycline**). **SABE** (subacute bacterial endocarditis; *Strep. viridans* & *faecalis*): with penicillin — but **gentamicin preferred**. **Reserve first-line TB drug**, only in combination.
Gentamicin	From *Micromonospora purpurea*. **Most commonly used** aminoglycoside. More potent than streptomycin; broader spectrum (Pseudomonas, Proteus, E. coli, Klebsiella, Enterobacter, Serratia). Low cost, reliable, long experience. **Synergistic** with ampicillin, penicillin G, ticarcillin, ceftriaxone, vancomycin. **Ineffective against *M. tuberculosis*.** Relatively **more nephrotoxic**.
Gentamicin — uses	Restricted to **serious Gram –ve bacillary infections**: septicaemia, sepsis, fever in **immunocompromised** (with penicillins) · **pelvic infections** (+ metronidazole) · **SABE** (+ Pen G or ampicillin or vancomycin) · **coliform infection** (+ ampicillin or ceftriaxone) · **Pseudomonal** infections (+ ticarcillin) · **Gram –ve meningitis**: 3rd-gen cephalosporin alone or with gentamicin.
Tobramycin	Identical to gentamicin; used in **Pseudomonas and Proteus**; oto- and nephrotoxicity **probably lower**.
Sisomicin	Identical to gentamicin; **more potent** on **Pseudomonas** and **β-hemolytic streptococci**; used interchangeably with gentamicin.
Amikacin	**Less toxic semisynthetic derivative of kanamycin**. **Resistant to enzymes** that inactivate gentamicin & tobramycin. **Widest spectrum**. Uses: same as gentamicin; **reserve drug for hospital-acquired Gram –ve bacillary infections**; **multidrug-resistant TB** (with other drugs). **Dose 15 mg/kg/day in 1–3 doses.**
Netilmicin	Semisynthetic derivative of **sisomicin**. Relatively resistant to aminoglycoside-inactivating enzymes. **More** active vs **Klebsiella, Enterobacter & staphylococci**; **less** active vs *Pseudomonas aeruginosa*. Doses and PK similar to gentamicin.
Neomycin	Wide spectrum vs Gram –ve bacilli and some Gram +ve cocci; **Pseudomonas and Strep. pyogenes NOT sensitive**. **Too toxic for parenteral use → topical/oral only.** **Uses:** topical (skin, eye, external-ear infections; combined with **bacitracin or polymyxin B** to widen spectrum and prevent resistant strains) · **orally**: **bowel preparation before surgery, 1 g TDS**; **hepatic coma** (suppresses ammonia-forming coliforms → prevents encephalopathy; **lactulose more preferred**) · **bladder irrigation** with polymyxin B.
Framycetin	Very similar to neomycin; too toxic for systemic use; topical on skin, eye, ear.

أهم ما يُسأل عنه

- **السمّيات الثلاث:** الأذن (VIII)، الكلية، العضلة (NM blockade).
- **Vestibular** ← streptomycin, gentamicin | **Cochlear** ← neomycin, amikacin, kanamycin.
- **Tobramycin**: أقل خطرًا على NM junction، ولها نوعا السمية السمعية.
- **Amikacin** أوسع طيف، ولا تدمّره الإنزيمات التي تحطم gentamicin؛ ويُستخدم في **MDR-TB**.
- **Neomycin** لا يُحقن (سمّي جدًا)؛ يُستعمل موضعيًا أو فمويًا (تحضير الأمعاء، hepatic coma).
- **Gentamicin** لا يعمل على السل ولا على اللاهوائيات؛ ويعطي **synergy** مع β-lactams.
- الجرعة في الفشل الكلوي: قلل الجرعة أو أطِل الفاصل (**q48h** عند CrCl أقل من 20).

Lecture notes: The slides write "Netilimicin" and "gentamycin" (spelling variants). The slide says "↓ clearance of antibiotic → ↑ ototoxicity" under nephrotoxicity — keep it as written. The renal-dose table figures are exactly as in the slide image.

3 · Sulfonamides, Trimethoprim & Co-trimoxazole

Source: "Sulfonamides and Trimethoprim" (PH343)

3.1 Sulfonamides — Introduction

- Introduced in the **1930s**, in clinical use since **1935**. **Bacteriostatic, synthetic** antimicrobials.
- Usefulness is limited by: **bacterial resistance** · adverse effects (**crystalluria**) · availability of more effective antibiotics.
- Now largely superseded by newer antibiotics and by **co-trimoxazole** (synergistic activity).
- **A sulfonamide alone is not the drug of first choice for any bacterial pathogen.**

3.2 Mechanism of action — the folate pathway

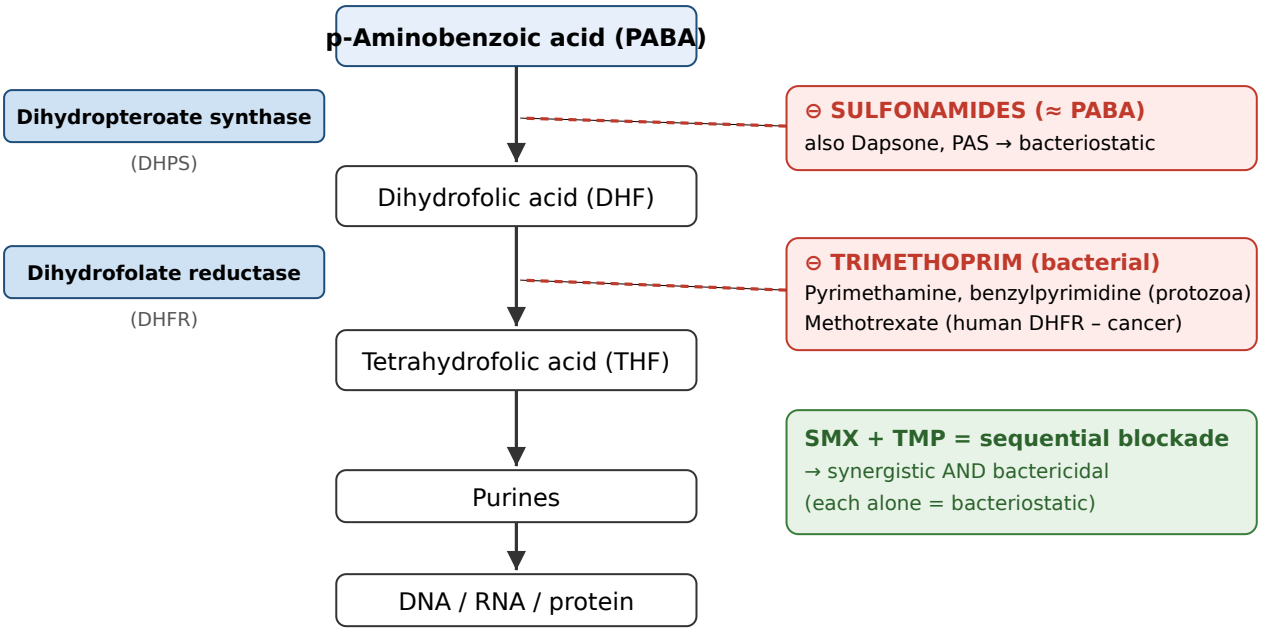


Figure 3.1 — bacterial folic-acid synthesis

- Certain microbes need **PABA** to form **dihydrofolic acid** — essential for purines and nucleic-acid synthesis.
- Sulfonamides are **structural analogs of PABA** and **competitively inhibit dihydropteroate synthase (DHPS)**; they form non-functional folic-acid analogues → stops **tetrahydrofolic acid** (coenzyme for DNA, RNA, protein synthesis).
- **Reversible** inhibitors → **bacteriostatic**. **Selective**: humans obtain folic acid from diet/exogenous sources rather than synthesizing it.

شرح

البكتيريا **تصنّع** الفولات بنفسها انطلاقًا من PABA، أما الإنسان فيأخذه من الطعام ← لهذا السلفا انتقائية السمّية. السلفا تشبه PABA فتنافسه على إنزيم **DHPS**. أما **Trimethoprim** فيمنع الخطوة التالية (إنزيم **DHFR**). الجمع بين الاثنين = حصار متتابع (sequential blockade) فيصبح التأثير **قاتلًا** بدل الإيقافي.

3.3 Antibacterial activity & resistance

Activity	Organisms
Primarily bacteriostatic vs many Gram +ve and Gram –ve	*Nocardia* · *Chlamydia trachomatis* · some protozoa (**Toxoplasma**) · some enterics: *E. coli, Klebsiella, Salmonella, Shigella, Enterobacter*
Now RESISTANT (formerly susceptible)	**Meningococci, pneumococci, streptococci, staphylococci, gonococci**
Rickettsiae	**NOT inhibited — actually stimulated in growth**

Resistance — mutations causing:

- ↑ production of PABA · **low affinity of DHPS (folate synthetase)** · alternative metabolic pathway for folate · loss of permeability / efflux of drug · **cross-resistance**.

3.4 Classification & pharmacokinetics

Group	Drugs
1. Oral absorbable — Short-acting	Sulfisoxazole, Sulfamethizole (t½ **6–9 h**)
— Medium-acting	Sulfamethoxazole, Sulfapyridine, Sulfadiazine (t½ **10–17 h**)
— Long-acting	**Sulfadoxine** (t½ **7–9 days**); slow urinary excretion, **high protein binding (>85 %)**, extensive tubular re-absorption of free drug
2. Oral non-absorbable	**Sulfasalazine** — local effect in gut, not absorbed, excreted largely in **feces**
3. Topical	**Sulfacetamide**, **Silver sulfadiazine**

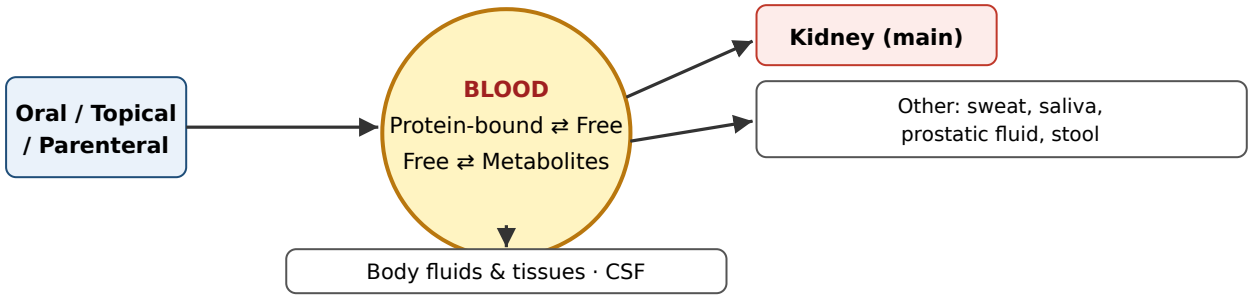


Figure 3.2 — sulfonamide disposition

- **Rapid absorption** from GIT; widely distributed into tissues and body fluids; **crosses placenta and BBB (CSF)**.
- Plasma protein binding variable **20–90 %**.
- **Metabolism** in liver by **acetylation or glucuronidation** → metabolites are **inactive, toxic and less soluble**.
- **Excretion:** primarily renal (free drug + metabolites). **Reduce dose in renal impairment.** Some are **very insoluble in acid urine** (→ crystalluria).

3.5 Clinical uses of sulfonamides

- Less used. Primary care only when **first-line** antibiotics were ineffective or contraindicated. **Replaced by co-trimoxazole in hospital**.

Group	Uses
Oral absorbable	• **Toxoplasmosis** — **sulfadiazine + pyrimethamine**: first line, treatment and prevention of recurrence; give **folinic acid 10 mg orally daily** to minimize bone-marrow depression. • **Malaria** — **sulfadoxine + pyrimethamine (Fansidar)** for **chloroquine-resistant *falciparum***. • **UTI** — sulfisoxazole, sulfamethoxazole. • **Nocardiosis**. • Respiratory infection, sinusitis, pneumonia, otitis media and dysentery in developing countries: **NOT RECOMMENDED**.
Oral non-absorbable	**Sulfasalazine**: **ulcerative colitis**, enteritis, other IBD. Split by intestinal flora into **sulfapyridine + 5-aminosalicylic acid (5-ASA)**; the salicylate is released at high local concentration (not achievable orally) and gives the **anti-inflammatory effect**. **Olsalazine**: a **dimer of salicylic acid** split by colonic bacteria; active in ulcerative colitis.
Topical	**Sulfacetamide**: ocular infections — **trachoma** and bacterial **conjunctivitis**. **Silver sulfadiazine 1 % cream**: prevention of infection of **burn wounds** (superficial and mid-dermal) — **no longer recommended**, though still an effective treatment; replaced by newer **occlusive dressings** (faster healing, less pain, fewer dressing changes, better satisfaction).

3.6 Adverse effects of sulfonamides

- Relatively common — about **3 %** of people on a course. Older people at higher risk.
- **Common:** GIT (nausea, vomiting, anorexia, diarrhea) and **hyperkalaemia**; hypersensitivity (rash, **Stevens-Johnson syndrome**).
- **Severe: hemolytic anemia in G6PD deficiency**; **crystalluria** (prevent by hydration + urine alkalinization); **kernicterus in neonates** (displacement of bilirubin).

Type	Details
1. Hypersensitivity	Occurs in **0.09 %** (higher in **AIDS** patients); develops **1–3 weeks** after starting, resolves **1–2 weeks** after stopping; if it occurs **avoid sulfonamide antibiotics**. Fever, skin rashes, **photosensitivity**, exfoliative dermatitis, urticaria; also anaphylaxis, **toxic epidermal necrolysis**, hemolytic anemia, angioedema, thrombocytopenia, pancytopaenia, **Stevens-Johnson**, lupus-like syndrome, multi-organ hypersensitivity, pneumonitis, hepatitis, **interstitial nephritis**, systemic vasculitis.
2. Urinary tract	Precipitate in urine, especially at **neutral/acid pH → crystalluria, hematuria, obstruction**. Rare with soluble ones (e.g. **sulfisoxazole**). Treat: **sodium bicarbonate** (alkalinize urine) + fluids for hydration. Also nephrosis and allergic nephritis.
3. Hematopoietic	Hemolytic anemia, **aplastic anemia**, agranulocytopenia, thrombocytopenia, leukemia; **hemolysis in G6PD deficiency**.
4. Kernicterus	Newborns if taken by mother near the **end of pregnancy**: sulfonamides **displace bilirubin** from plasma-protein binding sites → free (lipid-soluble) bilirubin enters CNS → high brain concentrations → kernicterus.

Contraindications: **Newborn** · **Pregnancy** — avoid in **first trimester** (folate essential) and **after 32 weeks** (↑ risk of neonatal haemolysis and methaemoglobinaemia) · **Allergy** to the drug.

3.7 Trimethoprim & Co-trimoxazole (TMP-SMX)

- **Co-trimoxazole** = **trimethoprim + sulfamethoxazole**, two agents acting **synergistically**; the most widely used sulfonamide-TMP combination.
- **Three drugs inhibit dihydrofolate reductase: Trimethoprim** (selective for the **bacterial** enzyme) · **Pyrimethamine, benzylpyrimidine** (protozoal) · **Methotrexate** (**human** enzyme — cancer therapy).
- **Fixed-dose combination**, similar half-life (**≈10 h**); **optimal ratio 5 : 1** (SMX 400 mg : TMP 80 mg).
- SMX inhibits dihydropteroate synthase (↓ DHF); TMP inhibits DHFR (↓ THF → no purines/DNA) → **sequential blockade → bactericidal** (vs bacteriostatic sulfonamide alone).

Advantages of the combination
Expanded number of organisms inhibited · **Bactericidal** · **Synergism** · **Decreased resistance** · **Decreased toxicity**

Pharmacokinetics (TMP and TMP-SMX)

- TMP usually **oral**; TMP-SMX **oral and IV**.
- TMP: **well absorbed**, widely distributed incl. **CSF**. More soluble than SMX → **larger volume of distribution**.
- Drugs/metabolites excreted in urine within **24 h**. **Halve the dose if CrCl < 30 ml/min.**
- Lecture states TMP is a **"weak acid"** and **concentrates in prostatic and vaginal fluids** → more antibacterial activity there.

Clinical uses

Form	Uses
1. Oral trimethoprim	**Acute UTI**
2. Co-trimoxazole (first line, hospital only)	**Pneumocystis pneumonia** and **nocardiosis** in immunocompromised · toxoplasmosis (sulfadiazine + pyrimethamine) · **UTI and prostatitis** · GIT infections: **Shigella, Salmonella**
Co-trimoxazole (second line)	When first-line choices are ineffective/contraindicated or sensitivities indicate it: acute exacerbation of **chronic bronchitis** · **pneumonia in adults** · **otitis media in children** · **sinusitis** · **salmonellosis** (alternatives to β-lactams)
3. IV TMP-SMX	**Pneumocystis pneumonia — DOC** for moderately severe to severe infections in **HIV** patients · **Gram –ve sepsis** incl. resistant species (**Enterobacter, Serratia**), shigellosis, **typhoid fever**, UTI — when the patient can't take it orally
4. Oral pyrimethamine + sulfonamide	**Leishmaniasis and toxoplasmosis**: pyrimethamine + sulfadiazine · **Falciparum malaria**: pyrimethamine + sulfadoxine (**Fansidar**)
5. Prophylaxis	**HIV/AIDS-related opportunistic infections**

Adverse effects, interactions, contraindications of TMP / TMP-SMX

- Sulfonamide-related effects (above).
- **Trimethoprim** (antifolate properties): **megaloblastic anemia, leucopenia, agranulocytopenia** — avoided by co-administration of **folinic acid (leucovorin) 6–8 mg/day**; **hyperkalemia** (blocks epithelial sodium channels).
- **AIDS** patients treated for *Pneumocystis*: high frequency of reactions — **fever, rash, leucopenia, diarrhea, ↑ hepatic aminotransferases, hyperkalemia, hyponatremia**.
- **Interactions:** ↑ anticoagulant effect of **warfarin** · with **methotrexate** → enhanced toxicity.
- **Contraindications:** pregnancy and lactation (teratogenicity, kernicterus) · severe renal or hepatic impairment.

أهم ما يُسأل عنه

- السلفا = **PABA analog** تثبط (bacteriostatic) **DHPS**؛ TMP تثبط **DHFR**؛ والجمع = **bactericidal +** **synergistic**، النسبة **5:1**.
- **Co-trimoxazole**: DOC لـ **Pneumocystis pneumonia** (IV في مرضى HIV).
- **Malaria** = sulfadiazine + pyrimethamine (+ folinic acid). **Toxoplasmosis مقاوِم للكلوروكين** = Fansidar (sulfadoxine + pyrimethamine).
- **Sulfasalazine** → sulfapyridine + 5-ASA (الفعّال مضاد الالتهاب هو 5-ASA) للتهاب القولون التقرحي.
- **Silver sulfadiazine** للحروق؛ **Sulfacetamide** للعين (trachoma, conjunctivitis).
- ممنوعة في: **الحمل (أول ثلث وبعد الأسبوع 32)**، حديثو الولادة (**kernicterus**)، (hemolysis) **G6PD deficiency**.
- Crystalluria ← ترطيب + قلونة البول بـ **sodium bicarbonate**. Rickettsiae لا تتأثر بل **تتحفز**.
- جرعة TMP-SMX تُخفَّض للنصف إذا **CrCl أقل من 30**.

Lecture notes: (1) The slide describes trimethoprim as a **"weak acid"**; in standard pharmacology references TMP is a **weak base** (it concentrates in the more acidic prostatic fluid). If the exam is based on this lecture, know that the slide wording says "weak acid". (2) Typo in slides: "Chlamydia trichomatis" = *trachomatis*. (3) "Hyperkalaemia" is listed by the slide among sulfonamide GIT effects — as written. (4) Slide on silver sulfadiazine: "no longer recommended" yet "still an effective treatment".

4 · Broad-Spectrum Antibiotics

Source: "Broad spectrum antibiotics" — macrolides, lincosamides, fusidic acid, polymyxins, mupirocin, bacitracin, tetracyclines, chloramphenicol

Broad-spectrum antibiotics target a wide range of bacteria and treat infections caused by multiple bacterial strains.

4.1 Where the protein-synthesis inhibitors act

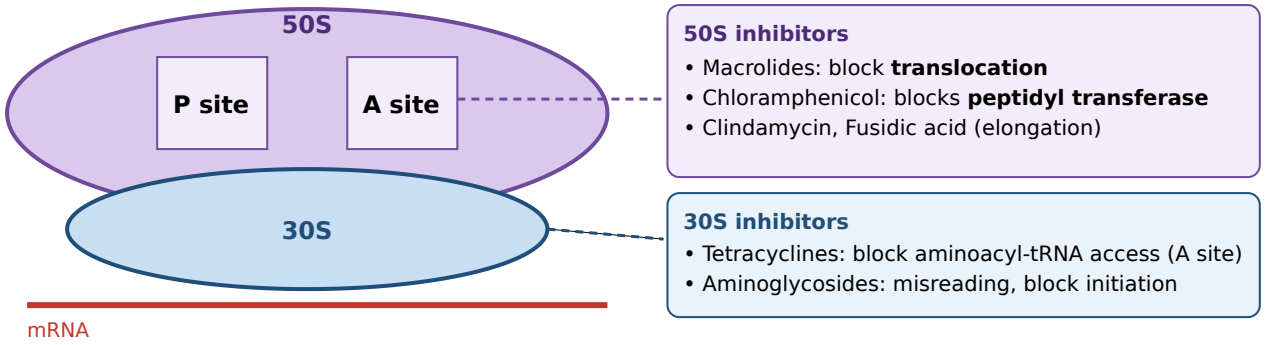


Figure 4.1 — site of action (based on lecture ribosome figures)

4.2 MACROLIDES

Erythromycin, Clarithromycin, Azithromycin, Roxithromycin — a **multi-membered lactone ring** with one or more deoxy sugars. **Prototype = erythromycin**; clarithromycin & azithromycin are **semisynthetic derivatives** of erythromycin.

- **MOA:** bind the **50S** ribosomal subunit → prevent **translocation** → inhibit protein synthesis. **Bactericidal or bacteriostatic** depending on drug concentration, bacterial type and medium pH.
- **Resistance:** reduced permeability of the cell membrane · active efflux · esterases that hydrolyze macrolides · modification of ribosomal binding site (chromosomal mutation).

Spectrum

Drug	Spectrum
Erythromycin	**Mainly Gram +ve:** *Staph. aureus, S. pneumoniae* (URTI e.g. otitis media, pharyngitis; LRTI e.g. pneumonia), *S. pyogenes, C. diphtheriae*. **Gram –ve:** *T. pallidum*. **Intracellular:** *L. pneumophila, M. pneumoniae, C. trachomatis*.
Clarithromycin	**Extends** the spectrum: *H. influenzae, H. pylori, M. catarrhalis, N. gonorrhoeae, M. avium*.
Azithromycin	**Higher activity** vs *H. influenzae, M. catarrhalis, C. trachomatis, M. pneumoniae, N. gonorrhoeae, Campylobacter jejuni, H. ducreyi*; also effective vs **Toxoplasma gondii**.

Pharmacokinetics

Drug	Points
Erythromycin	Absorption incomplete but adequate · **inactivated by gastric HCl → enteric-coated tablets** or esters (stearate, ethyl succinate) · food delays absorption · **not metabolized**; actively secreted in **bile** (major excretion route); only 2–5 % active in urine · widely distributed **except brain and CSF** · crosses placenta (**safe**) · protein binding 70–80 % · **t½ ≈ 1.6 h**
Clarithromycin	**Acid-stable** · food delays but doesn't alter extent of absorption · metabolized in liver to **14-hydroxy-clarithromycin (active)** · longer t½ → **twice daily**
Azithromycin	Some hepatic metabolism (inactive metabolites) · **biliary** elimination · **t½ ≈ 3 days** · **once daily** · **no inhibition of cytochrome P-450**

Clinical uses

1. **Alternative to penicillin** in allergic patients (staph, S. pyogenes, S. pneumoniae, T. pallidum) — erythromycin.
2. **Diphtheria & whooping cough** — erythromycin is the **drug of choice**.
3. Urogenital **Chlamydia in pregnancy** (safe). 4. **Syphilis** in penicillin-sensitive patients.
4. Urethritis in males / genital-tract infection in females due to **Chlamydia**.
5. **Mycoplasma pneumonia** (atypical) in children.
6. Azithromycin & clarithromycin: **toxoplasmosis encephalitis** and **H. pylori** (peptic ulcer); **atypical mycobacterial infections**; **H. influenzae** respiratory infection.
7. Prophylaxis of endocarditis (but **clindamycin replaced it**).
8. **Erythromycin + neomycin**: preoperative colon preparation. 12. Erythromycin as a **prokinetic** agent.

Adverse effects

- **GI disturbances** (erythromycin): macrolides are **motilin agonists** → nausea, vomiting, diarrhea, anorexia. Erythromycin causes severe GI distress, so **take with food**.
- **Liver toxicity: cholestatic jaundice**, especially with the **estolate** of erythromycin.
- **QT prolongation** — erythromycin has the highest tendency.
- Allergic reactions: urticaria, mild rashes.

Drug interactions

- Erythromycin metabolites **inhibit CYP450** → ↑ serum levels of **theophylline, oral anticoagulants, cyclosporine, methylprednisolone**.
- ↓ **digoxin** metabolism (inhibits gut flora that degrade digoxin) → ↑ systemic availability.

Comparison	Points
Clarithromycin vs erythromycin	1) more stable to gastric acid · 2) longer half-life · 3) **more active vs H. pylori, M. avium, toxoplasma** · 4) **lower GIT intolerance**
Azithromycin vs erythromycin	1) **less** active vs staph/strep, **more** active vs *H. influenzae* · 2) larger Vd and longer half-life · 3) **no drug interactions**

Telithromycin (Ketolide)

- Contains a **keto group** (lecture: "3 keto groups"); same spectrum as erythromycin but **less resistance** — structural modification makes it a **poor substrate for the efflux pump** and it binds ribosomes with **high affinity**. **Once daily.**
- Uses: community-acquired RTI (pneumonia, bronchitis, sinusitis, pharyngitis).
- Side effects: nausea & vomiting · **visual disturbance** · **may worsen myasthenia gravis** · QT prolongation.
- Interactions: ↑ toxicity of theophylline; potentiates warfarin, carbamazepine, astemizole, digoxin, cyclosporine.

4.3 LINCOSAMIDES — Clindamycin (Lincomycin: not used)

- **MOA:** bind **50S** → **bacteriostatic** inhibition of protein synthesis.
- **Spectrum:** Streptococci, pneumococci, staphylococci (**except MRSA**), ***Bacteroides fragilis***; **excellent vs *Corynebacterium (Propionibacterium) acnes***.
- **PK:** oral; distributes well to body fluids but **cannot penetrate the CNS**.
- **Uses:** bacteroides/anaerobic **abdominal, pelvic infection, peritonitis** (**not brain abscess**) · toxoplasmosis (with pyrimethamine) · **topical for acne** · **endocarditis prophylaxis** in valvular heart disease before dental procedure · bone & joint infections · dental infections.
- **Adverse effects:** most serious (fatal) — **pseudomembranous colitis** from overgrowth of *Clostridium difficile* toxins; skin rashes; impaired liver function.

4.4 Sodium fusidate (Fusidin)

- **MOA** (per lecture): inhibits protein synthesis by interfering with **elongation of the peptide chain**, binding the **50S** subunit.

- **Spectrum:** Gram +ve and Gram –ve **diplococci**; effective vs **β-lactamase-producing staph**. **Good penetration into bone and pus.**
- **Uses:** various infections, especially **osteomyelitis**. Given **orally, IV, and topically** (skin). Adverse effects rare except local irritation.

4.5 Polymyxin B & Colistin (Polymyxin E)

- **MOA:** peptide antibiotics act as **cationic detergents**; disrupt bacterial **cell-membrane** osmotic integrity by **displacing Ca^{2+} & Mg^{2+}** from membrane phospholipids → leakage of intracellular constituents → death.
- **Spectrum: bactericidal** on most Gram –ve bacilli **except Proteus, Serratia, Providencia & B. fragilis**.
- **Uses:** orally for oropharyngeal decontamination / suppression of intestinal flora in patients at high risk of endogenous infection; **colistimethate sodium inhaled** for respiratory infections, especially **cystic fibrosis**.
- **Side effects: neurotoxicity, nephrotoxicity, potent neuromuscular-blocking** properties.

4.6 Mupirocin & Bacitracin

Drug	Points
Mupirocin	From *Pseudomonas fluorescens*. Inhibits bacterial **RNA & protein synthesis** by binding to **isoleucyl-tRNA synthetase**. **Topical**: impetigo, folliculitis, burns, leg ulcers.
Bacitracin	**Bactericidal**; from *Bacillus subtilis*; Gram +ve cocci and bacilli, also *Neisseria* and *H. influenzae*. **MOA: inhibits cell-wall synthesis**. **Not absorbed orally; highly nephrotoxic parenterally → topical only**: infected wounds/ulcers, eye infections — **in combination with neomycin and polymyxin**.

4.7 TETRACYCLINES

Duration	Drugs
Short-acting	Chlortetracycline, Tetracycline, Oxytetracycline
Intermediate-acting	Demeclocycline, Methacycline
Long-acting	**Doxycycline, Minocycline**

- **Primarily bacteriostatic; broad spectrum** — aerobic & anaerobic, Gram +ve and –ve incl. **Rickettsiae, Chlamydiae, Mycoplasma, Legionella**; also some protozoa (**ameba**). More active vs Gram +ve than Gram –ve, but rarely indicated for Gram +ve infections (resistance, superior alternatives).
- **MOA:** bind **30S** → prevent access of **aminoacyl-tRNA to the acceptor (A) site** on the mRNA-ribosome complex → no new amino acids added to the peptide chain.

Pharmacokinetics

- **Absorption:** oral; old tetracyclines incompletely absorbed (low solubility; **chelate Ca^{2+}, Al^{3+}, Fe^{2+}, Mg^{2+}** in food/drugs). **Doxycycline and minocycline absorbed completely irrespective of food.**
- **Distribution:** wide (Vd >1 L/kg); concentrated in liver, kidney, spleen, connective tissue/bone; variable protein binding; **CSF level ≈ ¼ of plasma**.
- **Metabolism:** partly, with some **enterohepatic circulation**. **Excretion:** mainly **glomerular filtration** → dose reduced in renal failure.

Clinical uses

Infection	Notes
Rickettsial	**DOC**: Rocky Mountain spotted fever, **all forms of typhus**, **Q fever**
Mycoplasma	Tetracycline **or erythromycin** for *M. pneumoniae* pneumonia
Chlamydia	Lymphogranuloma venereum, chlamydial pneumonia, inclusion conjunctivitis, **trachoma**, non-specific urethritis
Bacillary	**Brucellosis, tularemia, cholera**
Others	**Acne**, exacerbation of bronchitis, community-acquired pneumonia · combination regimens for ***H. pylori*** ulcers · protozoal: *E. histolytica*, *P. falciparum*
Anthrax prophylaxis	**Doxycycline = DOC**
Spirochetal	**Lyme disease (*Borrelia burgdorferi*): Doxycycline = DOC**
SIADH	**Demeclocycline** blocks the effect of ADH → prevents water retention, ↑ serum sodium

Resistance

- Develops in graded manner · bacteria **pump the drug out** (efflux) · synthesis of a **protection protein** that shields the ribosomal binding site · **nearly complete cross-resistance** among tetracyclines.

Adverse effects

Effect	Details
1. Irritative	Epigastric pain, nausea, vomiting, diarrhea; **IM injection very painful**
2. Liver	**Fatty infiltration** of the liver and jaundice (occasionally)
3. Kidney	Prominent only with existing kidney disease. **Fanconi syndrome** from **outdated (expired) tetracycline** — degraded products (epitetracycline, anhydrotetracycline, epianhydrotetracycline)
4. Phototoxicity	Higher with **demeclocycline and doxycycline**
5. Teeth & bones	**Calcium–tetracycline chelate** deposits in developing teeth and bone: given from mid-pregnancy to **5 months** of extra-uterine life → **brown discoloration**, ill-formed teeth, more caries; late pregnancy or childhood → temporary suppression of bone growth
6. Anti-anabolic	↓ protein synthesis → catabolic effect
7. Diabetes insipidus	**Demeclocycline** antagonizes ADH → ↑ water loss (diuresis)
8. Vestibular toxicity	**Minocycline**: ataxia, vertigo

Precautions: Not in **pregnancy, lactation, children** · avoid with **diuretics** · caution in renal/hepatic insufficiency · **never use expired preparations** · do not mix with penicillin (inactivation) · **do not inject intrathecally**.

4.8 CHLORAMPHENICOL

- **Primarily bacteriostatic**, **bactericidal at high concentration**; broad spectrum.
- Highly active vs **Salmonella incl. *S. typhi***; more active than tetracyclines vs ***H. influenzae, B. pertussis, Klebsiella***; less active vs Gram +ve.
- **MOA:** potent inhibitor of microbial protein synthesis; binds **reversibly** to the **50S** subunit at the **peptidyl transferase** site → inhibits **transpeptidation**. Also inhibits **mitochondrial protein synthesis** in mammalian cells; **erythropoietic cells** particularly sensitive.

Pharmacokinetics

- **Rapidly and completely absorbed** from GIT. 50–60 % protein bound; very wide distribution (large Vd); **passes freely into serous cavities and across BBB**; crosses placenta; secreted in bile and milk.

- **Metabolism:** mainly **conjugated with glucuronic acid in liver**; plasma t½ **3–5 h**. **Excretion:** metabolites mainly in urine.

Adverse effects

Effect	Details
1. Bone-marrow depression	**Most important adverse effect.** Two types: (a) **dose-related toxic** effect — anemia, leukopenia, thrombocytopenia; (b) **idiosyncratic** response (not dose-related).
2. Gray baby syndrome	Neonates, especially **premature**, given relatively large doses. Begins **2–9 days** after starting: vomiting, flaccidity, **cyanosis**, respiratory irregularity, vasomotor collapse, hypothermia, abdominal distension, loose green stools, **pale gray color**. Cause: ↑ blood level from **inadequate glucuronide conjugation** + **inadequate excretion of unconjugated drug**.
3. GIT	Nausea, vomiting, diarrhea
4. Hypersensitivity	Rash, fever, angioneurotic edema — uncommon

Therapeutic uses & resistance

- **Uses: bacterial meningitis** · anaerobic or mixed infections · **rickettsial** diseases · **typhoid fever** · **topical for eye infections**.
- **Resistance** (graded): **R-plasmid-encoded acetyltransferase** inactivates the drug · decreased permeability · lower ribosomal affinity.

Precautions: never for **minor infections** or unidentified etiology · not for infections treatable by **safer antibiotics** · avoid repeated courses · **daily dose ≤ 2–3 g**.

أهم ما يُسأل عنه

- **Erythromycin** = DOC لـ **diphtheria & whooping cough**، وآمن في الحمل لعلاج Chlamydia، وأعلى macrolide في **QT prolongation** و**cholestatic jaundice (estolate)** وتثبيط CYP450.
- **Azithromycin**: t½ ثلاثة أيام، جرعة مرة يوميًا، **لا يثبط CYP450**. **Clarithromycin**: يعطي metabolite فعّال (14-hydroxy) ويفيد في H. pylori و M. avium.
- **Clindamycin**: خطر **pseudomembranous colitis** (C. difficile)، لا يدخل CNS، ومفيد في anaerobes (Bacteroides) وحب الشباب.
- **Tetracyclines**: تترسب في الأسنان والعظام (لا تُعطى للحامل/الطفل)، تتخلّب مع Ca/Al/Fe/Mg (لا تؤخذ مع الحليب أو antacids)، Doxycycline = DOC لـ Demeclocycline. **Lyme + anthrax prophylaxis** لـ **SIADH**.
- **Chloramphenicol**: **Gray baby** + **bone marrow depression** (أهم أعراضه)، يعبر BBB فيُستعمل في meningitis، وجرعته اليومية ≥ 2–3 g.
- **Polymyxin/Colistin**: يعطّل غشاء الخلية، **Nephro + Neuro toxic** + NM blockade، لا يؤثر على Proteus/Serratia.

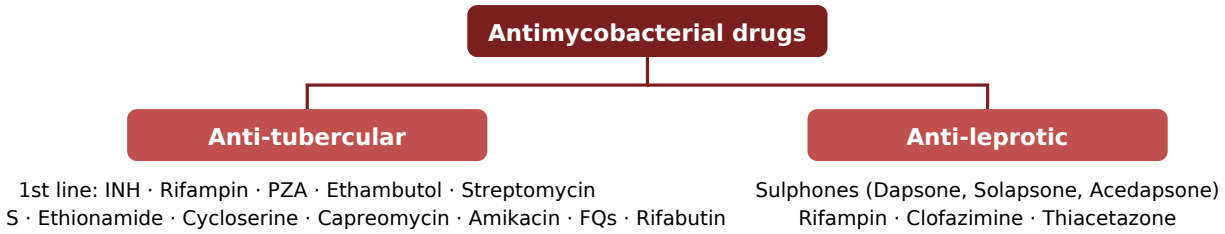
Lecture notes: (1) The slide's "Do not mix with penicillin in activation occurs" means **inactivation**. (2) "Plasm at 1/2" = plasma t½. (3) Slide's C. difficile line ("metronidazole or erythromycin") is ambiguous; the β-lactam lecture states **oral vancomycin first-line, metronidazole second-line** for C. difficile. (4) "Telithromycin contains 3 keto groups" — as written in the slide. (5) "Corynebacterium acnes" in the slide = *Propionibacterium (Cutibacterium) acnes*. (6) Slide says teeth effect from "mid-pregnancy to 5 months of extra-uterine life" — keep this wording for the exam.

5 · Antimycobacterial Drugs

Source: "Antimycobacterial Drugs" — Dr. Nabil Murghum, Pharmacology Department, Faculty of Medicine, Tripoli University

5.1 Mycobacteria & classification

- **Mycobacterium**: rod-like, **Gram-positive, aerobic** bacteria; some are pathogenic to humans.
- *M. tuberculosis* → **tuberculosis** | *M. leprae* → **leprosy** (Hansen's disease).



5.2 ANTITUBERCULAR DRUGS

Why TB treatment is complicated

- The bacteria **lack a peptidoglycan cell wall** (per lecture) and are **intracellular**.
- **Chronic** nature of disease · **development of resistance** · limited information on pharmacodynamics.
- **Always used in COMBINATION → delay resistance and augment efficacy.**

First-line drugs — pharmacokinetics

- **Well absorbed orally**, but absorption ↓ by **food (carbohydrates)** and **antacids Al(OH)₃**.
- Well distributed to all body fluids: **CSF level = plasma level**; they penetrate cells → effective on **intracellular** bacteria.
- **Metabolism:** Isoniazid — **acetylation** · Rifampin — **deacetylation** · Pyrazinamide — **deamination and oxidation**. Excreted mainly by the **kidney**.

First-line drugs — pharmacodynamics

Drug	Mechanism / activity
Isoniazid (INH)	Synthetic **analog of pyridoxine** (vit B6) · **prodrug** activated by a mycobacterial enzyme · inhibits **cell-wall synthesis** · **STATIC** on bacteria in **stationary** phase, **CIDAL** on **dividing** bacteria
Rifampin	Semisynthetic derivative of **rifamycin** · inhibits **RNA synthesis** · **CIDAL** vs *M. tuberculosis* and *M. leprae* · also **prophylaxis of meningitis due to H. influenzae** · treats **brucellosis** (with tetracycline)
Pyrazinamide (PZA)	Synthetic derivative of **nicotinamide** · **prodrug** activated by mycobacterial enzyme · inhibits cell-wall synthesis · **CIDAL**
Ethambutol	**STATIC**
Streptomycin	Aminoglycoside; was first-line but **no longer used** because: not absorbed orally (painful injection), **high toxicity**, resistance develops

Resistance mechanisms

Drug	Mechanism
Isoniazid	↓ accumulation of drug · alteration of target enzyme · ↑ amount of target enzyme
Rifampin	Change in bacterial permeability · **mutation in DNA-dependent RNA polymerase**

Side effects & interactions

Drug	Side effects
Isoniazid	**1. Neurotoxicity** (**slow acetylators**): **peripheral** — **paresthesia is the most common side effect** [INH → vit B6 deficiency → ↓ GABA synthesis → paresthesia]; **central** — restlessness, insomnia, convulsion, psychosis. **2. Hepatotoxicity** (**fast acetylators**) ↑ with: **old age, alcohol, concurrent rifampin**. **3. Hemolysis. 4. Hypersensitivity and SLE.** **Interaction:** inhibits **MAO-A** → after **tyramine-containing food**: flushing, palpitation, sweating, ↑ BP.
Rifampin	Flu-like syndrome · abdominal pain, nausea, vomiting · **hepatotoxicity**, hepatitis, jaundice · **harmless orange discoloration of body secretions**. **Interaction:** potent **enzyme inducer** → enhances clearance of many drugs.
Pyrazinamide	**Hyperuricemia and gout** · nausea, vomiting · rash and phototoxicity · dose-dependent **hepatotoxicity** · **hypoglycemia** (↑ glucose uptake by adipose tissue — as in slide)
Ethambutol	**Optic neuritis** → loss of visual acuity, **loss of red–green discrimination** · nausea, abdominal pain · peripheral neuropathy · rash, joint pain · hyperuricemia and gout · confusion, hallucination

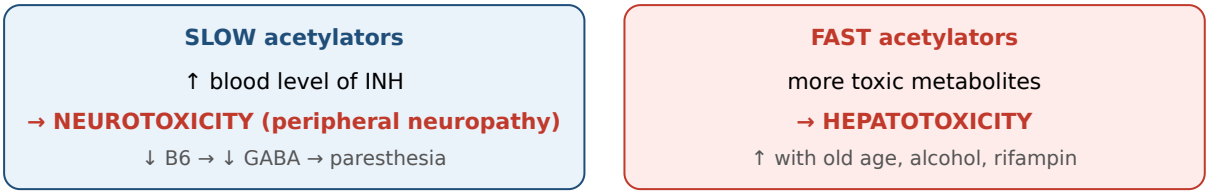


Figure 5.1 — isoniazid toxicity by acetylator type (as stated in the lecture)

Second-line drugs

- PAS (para-aminosalicylic acid) · Ethionamide · Cycloserine · Capreomycin · **Amikacin** · Fluoroquinolones · Rifabutin.
- **More toxic, less effective**; used **only when organisms are resistant** to first-line drugs.
- **PK:** well absorbed orally **except capreomycin** (poorly absorbed → given **IM**); widely distributed incl. CSF **except PAS (not to the brain)**; excreted mainly in urine.

Drug	Mechanism	Side effects / notes
PAS	**STATIC** — inhibits **folic-acid synthesis** → ↓ cellular growth	Gastric irritation · hepatitis · nephrotoxicity · **hypothyroidism** · hypersensitivity (rash, fever, joint pain)
Ethionamide	Inhibits cell-wall synthesis	Neuropathy · mental disturbances · nausea, gastric pain · **metallic taste**
Cycloserine	**STATIC** — inhibits **alanine racemase** → ↓ cell-wall synthesis	**Convulsions · psychosis · vertigo**. **Contraindicated in CNS diseases**
Capreomycin	**CIDAL**	**Ototoxicity, nephrotoxicity**
Amikacin	Aminoglycoside	Used as **second choice after streptomycin**

Regimens of TB therapy

Objectives: fast eradication · prevention of resistance.

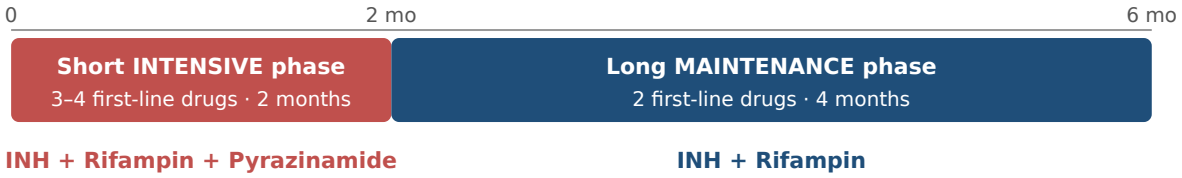


Figure 5.2 — standard short-course TB regimen

5.3 ANTILEPROTIC DRUGS

Classification: (i) **Sulphones** — Dapsone, Solapsone, Acedapsone · (ii) **Rifampin** · (iii) **Clofazimine** · (iv) **Thiacetazone**.

Drug	Details
Dapsone	**PK:** well absorbed orally; distributed throughout the body, **retained in skin, muscle, liver, kidney**; **t½ = 27 h**; metabolized by **acetylation** in liver; excreted in urine. **PD:** STATIC; acts **like sulfonamides** → inhibits folate synthesis → inhibits bacterial DNA synthesis. **Side effects: hemolytic anemia**, GIT irritation, skin rash, hepatitis, psychosis.
Solapsone	Less potent than dapsone; used in case of **gastric intolerance to dapsone**.
Acedapsone	**Long-acting** preparation for **IM** injection.
Rifampin	Given in **combination** to prevent resistance; **single monthly dose of 600 mg**.
Clofazimine	Well absorbed orally · **doesn't enter CNS** · eliminated in urine and bile · CIDAL · inhibits **bacterial DNA replication** · **anti-inflammatory** effect · causes **red-to-brown discoloration of skin and cornea** · GIT upsets.
Thiacetazone	Effective against *M. tuberculosis* and *M. leprae*; in leprosy used in case of **gastric intolerance to dapsone**.

Regimens of leprosy therapy

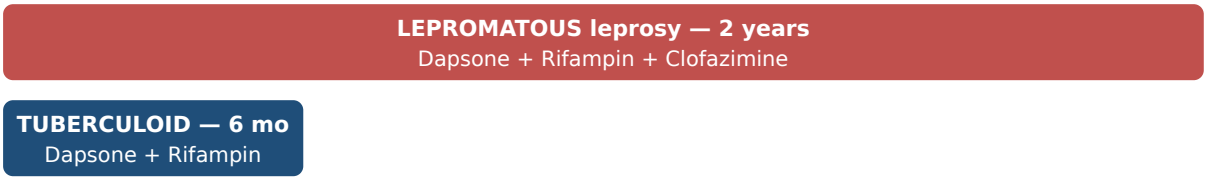


Figure 5.3 — leprosy regimens (objective: fast eradication + prevent resistance)

أهم ما يُسأل عنه

- **أدوية الخط الأول الخمسة:** INH, Rifampin, Pyrazinamide, Ethambutol, Streptomycin. تُعطى **دائمًا بالتركيب** (combination).
- **INH**: prodrug، analog لـ pyridoxine ← أهم أعراضه **paresthesia** (نقص B6)؛ **slow acetylators →** **neurotoxicity**، **fast acetylators → hepatotoxicity**؛ ويثبط **MAO** (تفاعل مع tyramine).
- **Rifampin**: يثبط **RNA polymerase**، **لون برتقالي** للإفرازات، **enzyme inducer**، وقاية من H. influenzae) meningitis).
- **Pyrazinamide**: **hyperuricemia/gout**. **Ethambutol**: **optic neuritis** (لا يميّز الأحمر والأخضر).
- **Cycloserine**: ممنوع في أمراض CNS. **Capreomycin**: oto + nephrotoxic. **PAS**: hypothyroidism. **Ethionamide**: metallic taste.
- العلاج: **2 شهر** (INH+Rif+PZA) ثم **4 أشهر** (INH+Rif).
- الجذام: 3 = **Lepromatous** أدوية لمدة **سنتين**؛ **Tuberculoid** = دواءان لمدة **6 أشهر**. **Clofazimine** = تلوّن الجلد والقرنية. **Dapsone** = hemolytic anemia.

(+) Mnemonic — First-line TB drugs "RIPES": **R**ifampin · **I**soniazid · **P**yrazinamide · **E**thambutol · **S**treptomycin.

Lecture notes: (1) The slide says the neurotoxic side effects occur in **slow** acetylators and hepatotoxicity in **fast** acetylators — keep this exactly for the exam. (2) The slide says streptomycin is "not absorbent orally" (= not absorbed orally). (3) The slide says "Causative bacteria don't have a peptidoglycan cell wall" — reproduced as written. (4) "Thiacetazone" and "Solapsone" are alternatives only when dapsone is not tolerated.

6 · Exam Cheat-Sheet & Self-Test

Everything high-yield in a few pages — review this the night before

6.1 Drug of choice / best-known indication

Infection / situation	Drug (as stated in the lectures)
Diphtheria & whooping cough	**Erythromycin** (DOC). Pen G only eliminates diphtheria *carrier state*; antitoxin = the effective treatment
Gonorrhea	**Ceftriaxone** 250 mg IM single dose (DOC)
Anthrax — treatment / prophylaxis	**Penicillin G** (DOC) / **Doxycycline** (DOC for prophylaxis)
Rat-bite fever	**Penicillin G** (DOC)
Lyme disease (Borrelia)	**Doxycycline** (DOC)
Rickettsial (Rocky Mountain spotted fever, typhus, Q fever)	**Tetracycline** (DOC)
Chlamydia in pregnancy · syphilis in penicillin-allergic	**Erythromycin**
Pneumocystis pneumonia (HIV, moderate–severe)	**IV TMP-SMX (co-trimoxazole)** (DOC)
Toxoplasmosis	**Sulfadiazine + pyrimethamine** (+ folinic acid) — first line; azithro/clarithro for encephalitis; clindamycin + pyrimethamine
Chloroquine-resistant *P. falciparum*	**Fansidar** = sulfadoxine + pyrimethamine
ESBL Gram –ve bacteria	**Carbapenems** (DOC)
MRSA (severe) · *C. difficile* colitis	**Vancomycin** IV · **Vancomycin PO** (metronidazole 2nd line)
Hospital-acquired Gram –ve bacillary (reserve)	**Amikacin**
Pseudomonas	Extended-spectrum penicillins (carbenicillin, ticarcillin, piperacillin…) · **ceftazidime** · cefepime · carbapenems · tobramycin/gentamicin (+ ticarcillin)
Penicillin/cephalosporin/carbapenem-allergic Gram –ve	**Aztreonam** (little cross-reactivity)
Burn-wound infection prevention · trachoma/conjunctivitis	**Silver sulfadiazine** (no longer recommended) · **Sulfacetamide**
Ulcerative colitis	**Sulfasalazine** (→ 5-ASA) · Olsalazine
SIADH	**Demeclocycline**
Hepatic coma · preoperative bowel prep	**Neomycin** PO (lactulose preferred in coma) · neomycin ± erythromycin
Impetigo / folliculitis · acne	**Mupirocin** topical · **clindamycin** topical (and tetracyclines)
Osteomyelitis	**Fusidic acid** (good bone penetration); clindamycin for bone & joint
Brucellosis · plague · tularemia	**Streptomycin + tetracycline** (brucellosis also rifampin + tetracycline)
Typhoid · bacterial meningitis · anaerobic/mixed	**Chloramphenicol** (also IV TMP-SMX / ampicillin for typhoid)
Meningitis in children · Gram –ve meningitis	Ampicillin + 3rd-gen cephalosporin · 3rd-gen cephalosporin ± gentamicin
Anaerobes · *Giardia* · *Trichomonas* · bacterial vaginosis	**Metronidazole** (clindamycin for *Bacteroides* abdominal/pelvic)
H. pylori	Clarithromycin / azithromycin · metronidazole / tetracycline in combination regimens
MDR-TB · Leprosy	Amikacin + other drugs · Dapsone + rifampin (± clofazimine)

6.2 "Signature" adverse effect → drug

Adverse effect	Drug(s)
Anaphylaxis (adrenaline)	Penicillins (cross-allergy ≈ 10 % with cephalosporins)
Hypokalemia	Carbenicillin
Seizures	Imipenem (renal failure) · cycloserine · INH (central) · high-dose penicillin (renal insufficiency)
Hypoprothrombinemia → vitamin K	Cephalosporins
Disulfiram-like reaction	Cephalosporins (cefoperazone) · **Metronidazole**
Red man syndrome	Vancomycin (rapid IV infusion)
Ototoxicity · nephrotoxicity · NM blockade	Aminoglycosides · vancomycin · polymyxins/colistin (neuro+nephro+NM) · capreomycin (oto+nephro)
Cochlear (hearing) vs Vestibular	Neomycin, amikacin, kanamycin vs streptomycin, gentamicin
Crystalluria · kernicterus · G6PD hemolysis · Stevens-Johnson	Sulfonamides (also dapsone: hemolytic anemia)
Megaloblastic anemia, hyperkalemia	Trimethoprim (folinic acid prevents the anemia)
QT prolongation · cholestatic jaundice (estolate)	Erythromycin (highest QT) · telithromycin QT
Pseudomembranous colitis (*C. difficile*)	Clindamycin · ampicillin
Teeth discoloration / bone growth ↓ · Fanconi (expired) · phototoxicity	Tetracyclines (phototoxicity: demeclocycline & doxycycline)
Nephrogenic diabetes insipidus	Demeclocycline
Vestibular toxicity (ataxia, vertigo)	Minocycline (also streptomycin/gentamicin)
Bone-marrow depression · Gray baby	Chloramphenicol
Paresthesia (↓B6) · MAO inhibition (tyramine) · hepatitis	Isoniazid
Orange secretions · enzyme inducer	Rifampin
Hyperuricemia/gout	Pyrazinamide · ethambutol
Optic neuritis (red–green)	Ethambutol
Hypothyroidism · metallic taste · psychosis/vertigo	PAS · ethionamide · cycloserine
Skin/cornea red-brown discoloration	Clofazimine
Worsens myasthenia gravis	Telithromycin · aminoglycosides

6.3 Mnemonic collection

Memory aids

• Aminoglycoside spectrum **PECKS**: Proteus/Pseudomonas · E. coli/Enterobacter · Klebsiella · Shigella. Not anaerobes (needs O_2 uptake).
• Aminoglycoside toxicity: **"3 O/N"** — **O**to (VIII), **N**ephro, **N**euromuscular.
• **(+) "AT 30, CELL at 50"** — Aminoglycosides & Tetracyclines at 30S; Chloramphenicol, Erythromycin, Lincosamides, Linezolid at 50S.
• TB first-line **RIPES**. INH → **neuropathy (slow)** / **hepatitis (fast)**; "**S**low = **S**ensory nerves".
• Sulfonamides contraindicated in **newborn** (kernicterus), **pregnancy**, **G6PD**.
• Rifampin = **R**NA polymerase, **R**ed-orange secretions, **R**aises (induces) enzymes.
• Ethambutol = **E**yes. Pyrazinamide = **P**ain (gout). Isoniazid = **I**nflamed nerves.
• Penicillins act on **growing** cells only — never with a static drug.

6.4 Quick self-test (cover the right column)

#	Question	Answer
1.	Enzyme that performs the last (cross-linking) step of cell-wall synthesis?	**Transpeptidase (= PBP)**
2.	Why can't penicillins be combined with a bacteriostatic drug?	**They kill only actively growing cells**
3.	Gene/protein for MRSA resistance?	**mecA → PBP2a**
4.	Porin lost in resistant *Pseudomonas*?	**OprD**
5.	Partners: clavulanate / sulbactam / tazobactam?	**Amoxicillin (or ticarcillin) / ampicillin / piperacillin**
6.	Pen G half-life and long-acting forms?	**≈30 min; procaine penicillin G, benzathine penicillin**
7.	Acid-stable natural penicillin given orally?	**Penicillin V**
8.	Which anti-staph penicillin is acid-labile and nephrotoxic?	**Methicillin**
9.	Penicillin causing hypokalemia?	**Carbenicillin**
10.	Treatment of immediate penicillin anaphylaxis?	**Adrenaline**
11.	Gonorrhea DOC & dose?	**Ceftriaxone 250 mg IM single dose**
12.	Cephalosporin allergy cross-reaction with penicillin?	**≈ 10 %**
13.	Cephalosporin bleeding — prevention?	**Vitamin K (hypoprothrombinemia)**
14.	2nd-gen cephalosporin useful in meningitis (crosses BBB)?	**Cefuroxime**
15.	Cephalosporins with anti-Pseudomonas activity?	**Ceftazidime (3rd), cefepime (4th)**
16.	Why is imipenem given with cilastatin?	**Renal dehydropeptidase inactivates imipenem**
17.	DOC for ESBL-producing Gram –ve?	**Carbapenems**
18.	Aztreonam spectrum & special advantage?	**Gram –ve aerobes only; safe in β-lactam allergy**
19.	Vancomycin — oral use? Red man syndrome cause/prevention?	**Only for *C. difficile*; rapid infusion → slow ~60 min + antihistamine**
20.	Metronidazole MOA and alcohol reaction?	**Free radicals → DNA strand breaks; disulfiram-like (inhibits aldehyde dehydrogenase)**
21.	Why are aminoglycosides inactive vs anaerobes?	**Uptake across cytoplasmic membrane is O_2-dependent**
22.	Aminoglycoside target & consequences?	**30S; blocks initiation, misreading of mRNA, polysome → monosome**
23.	Aminoglycoside killing pattern and best dosing?	**Concentration-dependent + PAE → single daily dose**
24.	Which cause cochlear vs vestibular damage?	**Neomycin, amikacin, kanamycin vs streptomycin, gentamicin**
25.	Aminoglycoside with the least NM blockade?	**Tobramycin**
26.	Why amikacin is a reserve drug?	**Resists inactivating enzymes; widest spectrum; hospital-acquired Gram –ve, MDR-TB**
27.	Gentamicin — not effective against?	***M. tuberculosis* (and anaerobes)**
28.	Neomycin — routes and uses?	**Topical/oral only; bowel prep (1 g TDS), hepatic coma (lactulose preferred)**
29.	Cockcroft-Gault for females?	**× 0.85 [CrCl = (140−age)×wt ÷ (sCr×72)]**
30.	Dose/frequency at CrCl 20 ml/min?	**80 % every 48 h**

31. Enzyme blocked by sulfonamides / trimethoprim?	**DHPS / DHFR**
32. Co-trimoxazole ratio and effect?	**SMX:TMP = 5:1 (400:80); synergistic, bactericidal**
33. Why sulfonamides are selectively toxic to bacteria?	**Humans take folate from the diet**
34. Crystalluria — prevention/treatment?	**Hydration + sodium bicarbonate (alkalinize urine)**
35. Mechanism of kernicterus with sulfa?	**Displace bilirubin from albumin → free bilirubin enters CNS**
36. When to avoid sulfonamides in pregnancy?	**First trimester and after 32 weeks**
37. What are Fansidar's components and use?	**Sulfadoxine + pyrimethamine; chloroquine-resistant *falciparum***
38. Sulfasalazine splits into? Which is active?	**Sulfapyridine + 5-ASA; 5-ASA (anti-inflammatory)**
39. Effect of sulfonamides on Rickettsiae?	**Stimulate their growth**
40. TMP adverse effects and prevention?	**Megaloblastic anemia/leucopenia (folinic acid 6–8 mg/d), hyperkalemia**
41. Macrolide target and block?	**50S; prevents translocation**
42. Macrolide with highest QT risk & CYP inhibition?	**Erythromycin**
43. Azithromycin: t½, dosing, CYP?	**≈3 days, once daily, no CYP450 inhibition**
44. DOC for diphtheria / whooping cough?	**Erythromycin**
45. Clindamycin — most serious toxicity?	**Pseudomembranous colitis — *C. difficile***
46. Tetracycline target; why avoid with milk/antacids?	**30S (blocks aminoacyl-tRNA at A site); chelation with Ca, Al, Fe, Mg**
47. Tetracyclines absorbed irrespective of food?	**Doxycycline and minocycline**
48. DOC: Lyme disease, anthrax prophylaxis, SIADH?	**Doxycycline, doxycycline, demeclocycline**
49. Chloramphenicol target, major toxicities?	**50S (peptidyl transferase); bone-marrow depression, gray baby**
50. Cause of gray baby syndrome?	**Inadequate glucuronidation + inadequate excretion → ↑ drug level**
51. Polymyxin/colistin MOA & toxicity?	**Cationic detergent (displaces Ca/Mg); neuro-, nephrotoxic, NM blockade**
52. Mupirocin target?	**Isoleucyl-tRNA synthetase (topical, impetigo)**
53. Bacitracin route and combination?	**Topical only; with neomycin + polymyxin**
54. Fusidic acid — best use?	**Osteomyelitis (bone penetration), β-lactamase staph**
55. Why is INH called a prodrug and an analog of what?	**Activated by mycobacterial enzyme; pyridoxine analog**
56. INH slow vs fast acetylator toxicity?	**Slow → neurotoxicity; fast → hepatotoxicity**
57. Why does INH cause paresthesia?	**Vit B6 deficiency → ↓ GABA synthesis**
58. Drug that is an enzyme inducer, orange secretions?	**Rifampin**
59. TB drug with hyperuricemia / optic neuritis?	**Pyrazinamide / ethambutol**
60. Why streptomycin is no longer a first-line TB drug?	**Not orally absorbed (painful injection), toxic, resistance**
61. Second-line TB drug contraindicated in CNS disease?	**Cycloserine**

62. Second-line drug causing hypothyroidism / metallic taste?	**PAS / ethionamide**
63. Standard TB regimen?	**2 mo INH+Rif+PZA, then 4 mo INH+Rif**
64. Lepromatous vs tuberculoid leprosy regimen?	**Dapsone+Rif+Clofazimine ×2 y vs Dapsone+Rif ×6 mo**
65. Clofazimine hallmark side-effect?	**Red-brown skin & cornea discoloration**
66. Dapsone — MOA and toxicity?	**Like sulfonamides (folate); hemolytic anemia**